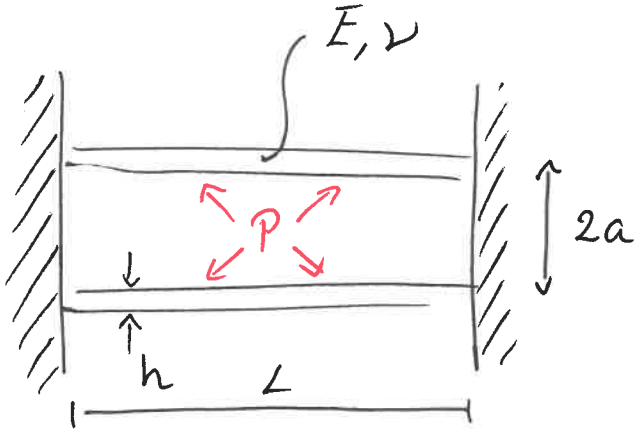
Övning 15 - Sammansatta tillstånd

$$S_{tot} = S_{\underline{I}} + S_{\underline{II}} + S_{\underline{III}} + \ldots$$

2.9.1



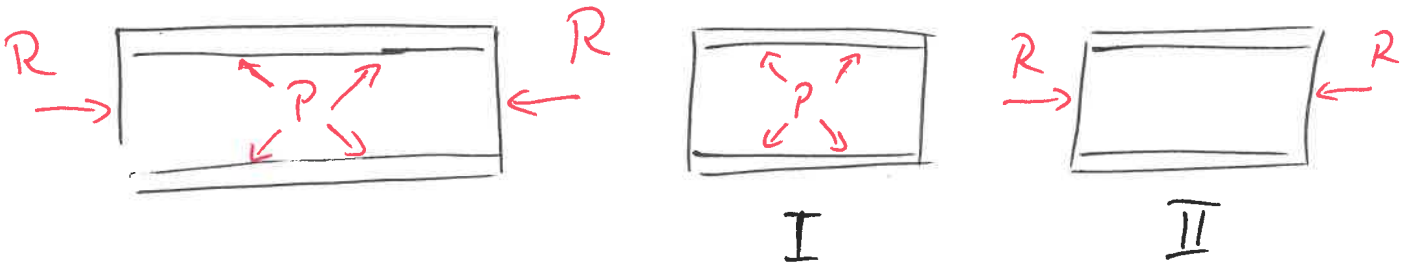
Sökt:

a) Reaktionskrafter från väggarna R

b) Huvudspänningar $\sigma_1\ \sigma_2\ \sigma_3$

① Frilägg

Superponera



Fast inspänd $\Rightarrow \delta_{tot} = \delta_I + \delta_{II} = 0$

(I) $\delta_I = \varepsilon_{zI} L \qquad \varepsilon_{zI} = \frac{1}{E}(\sigma_z - \nu(\sigma_r + \sigma_\varphi))$

[FS 7.31]

$\sigma_r \approx 0 \qquad \sigma_\varphi = \frac{Pa}{h} \qquad \sigma_z = \frac{Pa}{2h}$

$$\varepsilon_{zI} = \frac{Pa}{Eh}\left(\frac{1}{2} - \nu\right)$$

$$\boxed{\delta_I = \frac{PaL}{Eh}\left(\frac{1}{2} - \nu\right)}$$

2.9.1 del 2

(II) $\delta_{II} = \varepsilon_{z II} L \qquad \varepsilon_{z II} = \frac{N}{EA} = \frac{-R}{2a\pi hE}$

$$\boxed{\delta_{II} = -\frac{RL}{2a\pi hE}} \qquad \left(\sigma_z = -\frac{R}{2a\pi h} \quad \sigma_r = \sigma_\varphi = \tau_{r\varphi} \ldots = 0\right)$$

$$\left(\delta_I + \delta_{II} = 0\right) \Rightarrow \frac{PaL}{Eh}\left(\frac{1}{2} - \nu\right) - \frac{RL}{2a\pi hE} = 0$$

$$\Longrightarrow \boxed{R = Pa^2\pi(1-2\nu)}$$

b) Superponera spänningstillstånd

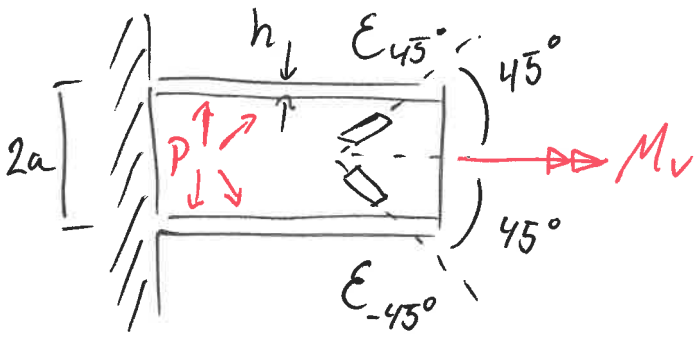
$$\boxed{\sigma_r = 0} \qquad \boxed{\sigma_\varphi = \frac{Pa}{h}} \qquad \sigma_z = \frac{Pa}{2h} - \frac{R}{2a\pi h}$$

$$\sigma_z = \frac{Pa}{2h} - \frac{Pa^2\pi(1-2\nu)}{2a\pi h} = \boxed{\nu\frac{Pa}{h}}$$

Inga skjurspänningar!

$$\sigma_1 = \frac{Pa}{h} \qquad \sigma_2 = \nu\frac{Pa}{h} \qquad \sigma_3 = 0 \quad (\text{om } P>0)$$

2.9.6



Tunnväggigt $h << a$

$a = 25\,mm$

$h = 2\,mm$

$E = 200\,GPa$

$\nu = 0.3$

$\varepsilon_{45°} = 5 \cdot 10^{-5}$

$\varepsilon_{-45°} = 2 \cdot 10^{-5}$

Sökt: P, M_v så de uppmätta töjningarna uppnås.

① Superponera och bestäm spänningar



I II

(I) $\sigma_r = 0$ $\sigma_\varphi = \frac{Pa}{h}$ $\sigma_z = \frac{Pa}{2h}$

(II) $\tau_{z\varphi,max} = \frac{M_v}{W_v}$ [FS 6.76] $W_v = 2\pi a^2 h$ [FS 6.81]

$\tau_{z\varphi,max} = \frac{M_v}{2\pi a^2 h}$

$$S = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{Pa}{h} & \frac{M_v}{2\pi a^2 h} \\ 0 & \frac{M_v}{2\pi a^2 h} & \frac{Pa}{2h} \end{bmatrix}_{r\varphi z}$$

2.9.6 del 2

Hookes lag => [FS 3.4]

$$\varepsilon_\varphi = \frac{Pa}{Eh}\left(1 - \frac{\nu}{2}\right)$$

$$\varepsilon_r = -\frac{3\nu Pa}{2Eh}$$

$$\varepsilon_z = \frac{Pa}{Eh}\left(\frac{1}{2} - \nu\right)$$

$$\gamma_{z\varphi} = \frac{M_v(1-\nu)}{\pi a^2 h E}$$

$\varepsilon(45)$ och $\varepsilon(-45)$ hända i φz-planet

$$\varepsilon(45) = \varepsilon_{45} = \frac{1}{2}\varepsilon_\varphi + \frac{1}{2}\varepsilon_z - \frac{1}{2}\gamma_{z\varphi} = 5\cdot 10^{-5}$$

$$\varepsilon(-45) = \varepsilon_{-45} = \frac{1}{2}\varepsilon_\varphi + \frac{1}{2}\varepsilon_z + \frac{1}{2}\gamma_{z\varphi} = 2\cdot 10^{-5}$$

2 ekv - 2 obekanta

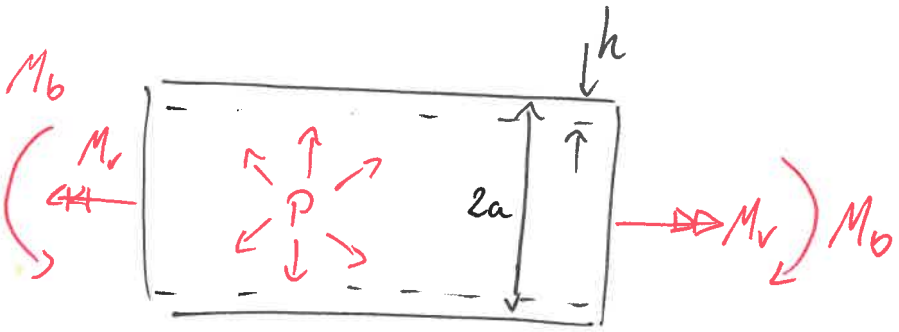
$$\Rightarrow \quad P = \frac{2(\varepsilon_{45} + \varepsilon_{-45})Eh}{3a(1-\nu)}$$

$$M_v = \frac{\pi a^2}{1+\nu}\left(2\varepsilon_{-45}Eh - Pa\left(\frac{3}{2} - \frac{3\nu}{2}\right)\right)$$

$$P = 4.6\cdot 10^5 \text{ Pa}$$

$$M_v = -42.3 \text{ Nm}$$

2.9.9



$a = 20\,mm$

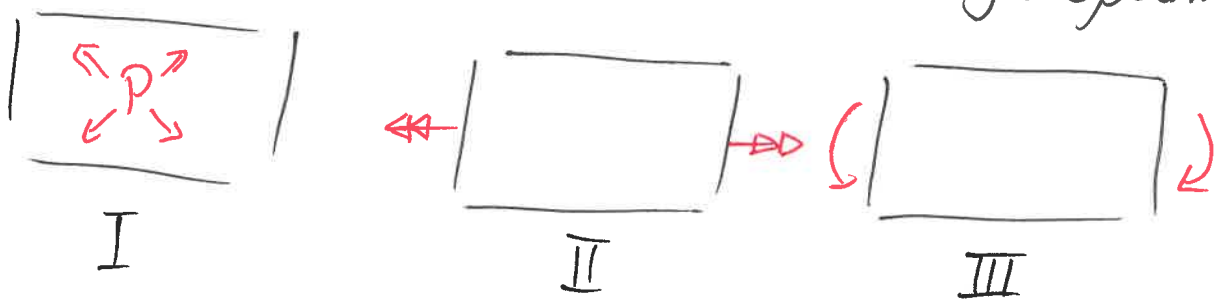
$h = 2\,mm$

$M_b = 1000\,Nm$

$M_v = 500\,Nm$

$p = 10\,MPa$

Sökt: Spänningstillståndet och max skjuvspänning



(I) $h = \frac{a}{10}$ tunnväggigt!

[FS 7.31] $\sigma_\varphi = \frac{pa}{h}$ $\sigma_z = \frac{pa}{2h}$ $\sigma_r \approx 0$ $\tau_i = 0$

(II) $\tau_{\varphi z,max} = \frac{M_v}{W_v}$ [FS 6.76]

$W_v = 2\pi a^3 t$ [FS 6.81]

$\tau_{\varphi z,max} = \frac{M_v}{2\pi a^3 h}$

2.9.9 del 2

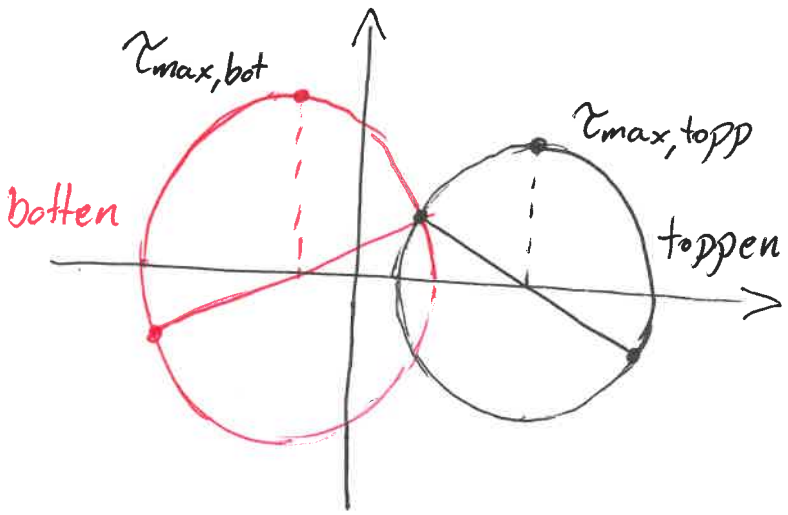
(III) $$\sigma_z = \frac{M_b}{I} y \quad [FS\ 6.7]$$

$$I = \pi a^3 t \quad [FS\ 30.1.4]$$

$$\sigma_{z,top} = \frac{M_b}{\pi a^3 h} a \qquad \sigma_{z,bot} = \frac{M_b}{\pi a^3 h} (-a)$$

$$S = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{Pa}{h} & \frac{M_v}{2\pi a^3 h} \\ 0 & \frac{M_v}{2\pi a^3 h} & \frac{Pa}{2h} \pm \frac{M_b}{\pi a^2 h} \end{bmatrix}$$

$$S_{toppen} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 100 & 99.4 \\ 0 & 99.4 & 448 \end{bmatrix} MPa \qquad S_{botten} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 100 & 99.4 \\ 0 & 99.4 & -298 \end{bmatrix} MPa$$

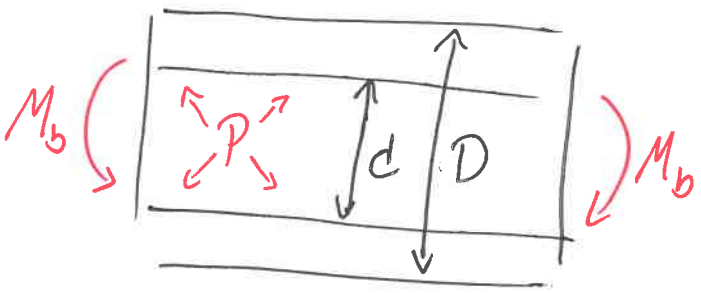


Botten

$$\tau_{max,bot} = R = \left(\left(\frac{\sigma_y - \sigma_z}{2} \right)^2 + \tau_{yz}^2 \right)^{\frac{1}{2}}$$

$$\boxed{\tau_{max,bot} = 245\ MPa}$$

2.9.10



Tjockväggigt rör

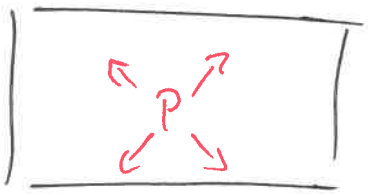
$D = 40\,mm$

$d = 20\,mm$

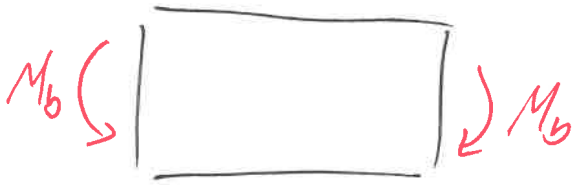
$P = 50\,MPa$

$M_b = 100\,Nm$

Sökt: Spänningstillstånd yttre och inre randen



(I)



(II)

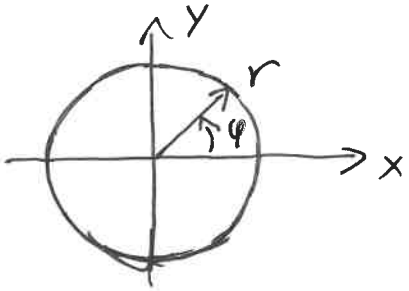
(I) $$\sigma_r = \frac{Pa^2}{b^2-a^2}\left(1-\frac{b^2}{r^2}\right)$$

$$\sigma_\varphi = \frac{Pa^2}{b^2-a^2}\left(1+\frac{b^2}{r^2}\right)$$ [FS 7.30]

$$\sigma_z = 0$$

(II) $$\sigma_z = \frac{M_b}{I}y \qquad I = \frac{\pi b^4}{4} - \frac{\pi a^4}{4}$$

$$y = r\sin\varphi$$



2.9.10

Inre randen $r = a = \frac{d}{2}$ $\left(b = \frac{D}{2}\right)$

$$\sigma_r = \frac{Pa^2}{b^2 - a^2}\left(\frac{a^2 - b^2}{a^2}\right) = -P$$

$$\boxed{\sigma_r = -50\,MPa}$$

$$\sigma_\varphi = \frac{Pa^2}{b^2 - a^2}\left(\frac{a^2 + b^2}{a^2}\right) = \frac{P(a^2 + b^2)}{b^2 - a^2}$$

$$\boxed{\sigma_\varphi = 83.3\,MPa}$$

$$\sigma_z = \frac{4M_b}{\pi(b^4 - a^4)}\, a \sin\varphi$$

$$\boxed{\sigma_z = 8.5 \sin\varphi\,MPa}$$

Yttre randen

$$\sigma_r = \frac{Pa^2}{b^2 - a^2}\left(\frac{b^2 - b^2}{b^2}\right) = 0$$

$$\boxed{\sigma_r = 0\,MPa}$$

$$\sigma_\varphi = \frac{Pa^2}{b^2 - a^2}\left(\frac{2b^2}{b^2}\right) = \frac{2Pa^2}{b^2 - a^2}$$

$$\boxed{\sigma_\varphi = 33.3\,MPa}$$

$$\sigma_z = \frac{4M_b}{\pi(b^4 - a^4)}\, b \sin\varphi$$

$$\boxed{\sigma_z = 17.0 \sin\varphi\,MPa}$$